

MINERVA CryoCell P&ID — Colour & Legend Guide (v4)

Project: MINERVA CryoCell — SCK CEN (MYRRHA/MINERVA Phase 1)

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Standard: SCK CEN AD_01.16 (P&ID conventions)

Status: S2 — FOR ACCEPTANCE · RESTRICTED

1. Purpose

This guide explains the colour coding, line-weight conventions and legend tables used across the v4 MINERVA CryoCell P&ID set. It applies to all four sheets (QCELL =NA.PS01_PFB712 and RFCELL =NA.PS01_PFB713 , process + utilities).

The drawing set is designed to be read in **two complementary modes**:

- **Colour (STD) mode** — fluid/service class is identified primarily by line colour.
 - **Mono mode** — for black-and-white printing, class is identified by **inline pipe names** printed directly on the run, plus differentiated line weights and dash patterns. No information is lost when colour is removed.
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2. Fluid / Service Class Colours

Each process class has a dedicated colour, design temperature and design pressure. These are summarised in the on-sheet **Class Legend** table (layer 16_Legend).

Class	Service	Colour	Hex	Design Temp	Design Press
D	40 K helium supply	Red	#e00000	40 K	14 bar
A	4.5 K helium	Blue	#0033cc	4.5 K	3 bar
B	2 K helium	Cyan	#00a6bd	2 K	27 mbar
WATER	Cooling water (CW)	Green	#00a000	Ambient	4 bar
E	60 K helium shield	Olive	#8a8a00	60 K	13 bar
AIR	Instrument / plant air	Magenta	#c000c0	Ambient	6 bar (g)

The colour swatches in the interactive HTML viewer's layer panel match these hex values exactly, so on-screen toggling reflects the printed legend.

3. Mono-Mode Line Weights & Names

When colour is suppressed (mono toggle / black-and-white printing), classes are distinguished by:

1. **Inline pipe names** — the line identifier is printed on the pipe run at regular intervals so the reader never has to trace a line back to a colour key. Examples:
 - 40K PRIMARY (Class D supply header)
 - 2K0-4" -PAC1-BA (Class B, 4-inch, line tag PAC1-BA)
2. **Line weight** — primary process headers are drawn heavier than branch/utility runs.
3. **Dash pattern** — signal and pneumatic lines use distinct dashed/dotted patterns (see Signal Legend below).

This guarantees AD_01.16 legibility on a monochrome A3 plot.

4. Signal / Connection Legend

The **Signal Legend** table (also on layer 16_Legend) documents instrument and connection line styles:

Line style	Meaning
Solid heavy	Major process pipe
Solid light	Minor process / branch pipe
Long dash	Electrical signal
Dash-dot	Software / data link
Fine dotted	Pneumatic / capillary
Double slash break	Pipe class / spec break

5. ISA Instrument Bubbles & Tag Boxes

- Instrument bubbles follow ISA-5.1: circle (field), circle-in-square (DCS/shared), hexagon (computer/PLC), with a horizontal bar for panel-mounted devices.
 - Every valve and instrument tag is rendered inside a **white tag box** so the text stays legible where it overlaps pipes, equipment or hatching — in both colour and mono modes.
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6. Edge Terminal Points

Off-sheet connections terminate at the **drawing frame edge** with a cloud-style terminal point carrying the continuation reference (AD_01.10 style). Terminal points are colour-classed in STD mode and labelled with the line name in mono mode.

7. Toggling Colour in the Interactive Viewer

In `HTML_INTERACTIVE/index.html` :

- **Mono toggle** — switches the whole set between colour and monochrome rendering.
- **Style toggle** — switches between the filled/coloured presentation and a line-weight-only presentation.
- **Layer panel** — each class layer has a colour swatch matching the table in §2; toggle individual classes on/off.

See `INTERACTIVE_VIEWER_MANUAL.md` for full viewer instructions.